The Fundamental Theorem of Calculus

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Part 2: evaluating definite integrals

Evaluate each definite integral:

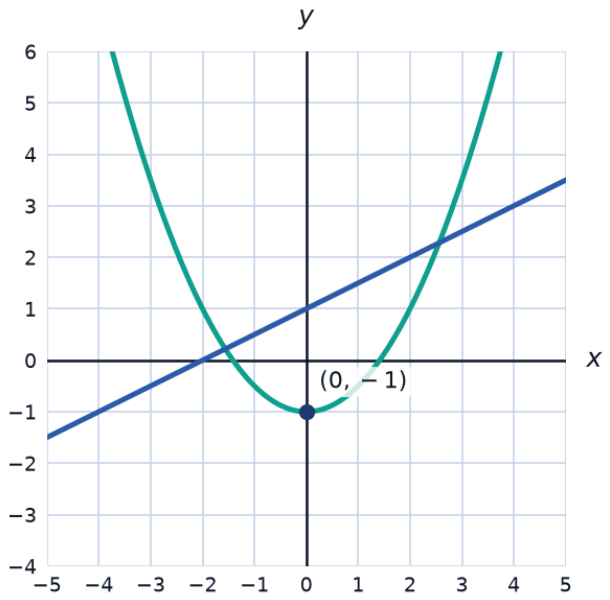
a $\int_1^3 (2x+1)\,dx$

b $\int_0^{\pi} \sin x\,dx$

c $\int_1^4 \sqrt{x}\,dx$

2 [Foundational] • Target: Part 2: evaluating definite integrals

Evaluate $\int_0^1 e^{2x}\,dx$, to three decimal places.



3 [Foundational] • Target: Part 1: differentiating an integral

Use the Fundamental Theorem, Part 1, to find $\frac{d}{dx}\int_2^x (t^3+1)\,dt$.

4 [Foundational] • Target: Part 1: differentiating an integral

Find $\frac{d}{dx}\int_1^{x^2} \sin(t)\,dt$ using the chain rule with FTC.

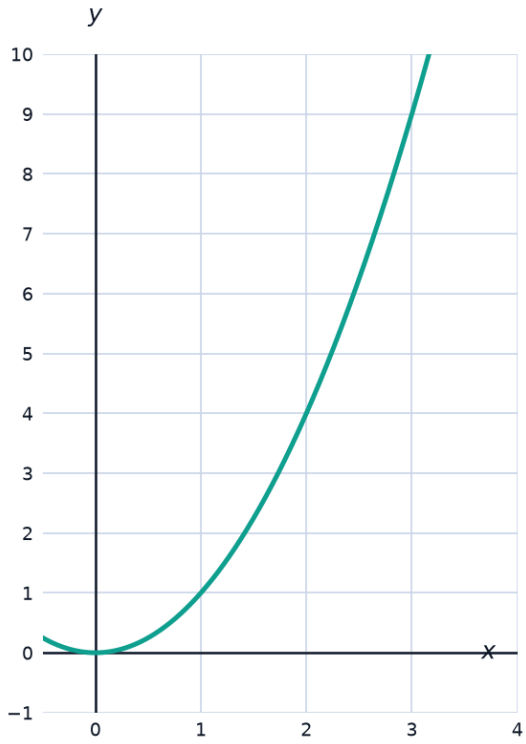
5 [Foundational] • Target: Part 1: differentiating an integral

Let $g(x) = \int_0^x \sqrt{4+t^2}\,dt$. Find $g'(3)$.

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6 [Foundational] • Target: Average value and applications

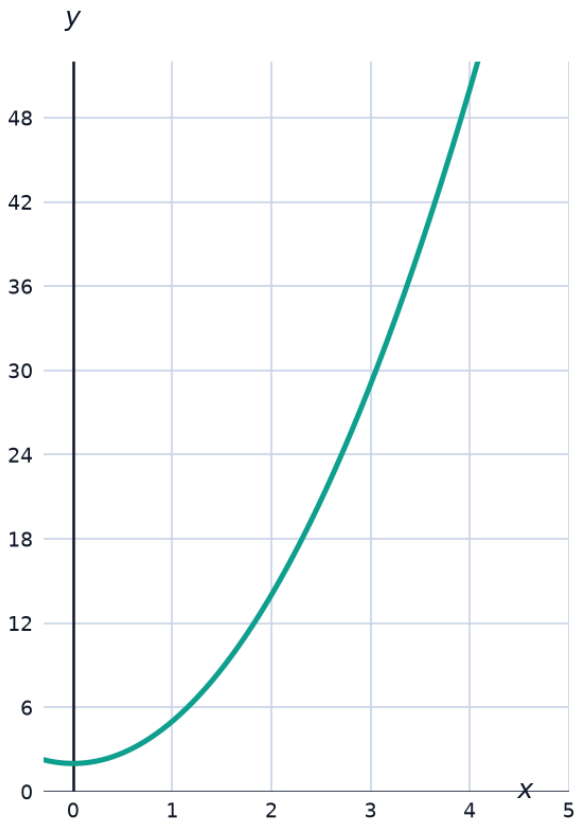
Find the average value of $f(x) = x^2$ on $[0, 3]$.



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7 [Foundational] • Target: Average value and applications

A car's velocity is $v(t) = 3t^2 + 2$ m/s for $0 \le t \le 4$. Find the total distance traveled, and the average velocity over this interval.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Further FTC practice

Evaluate $\int_0^2 (x^3 - 2x + 3)\,dx$.

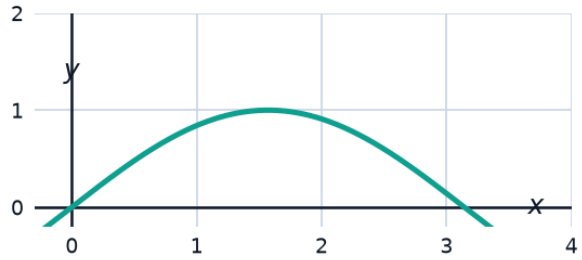
9 [Intermediate] • Target: Further FTC practice

Find $\frac{d}{dx}\int_x^5 \cos(t^2)\,dt$, using the property $\int_a^b = -\int_b^a$ before applying FTC Part 1.

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10 [Intermediate] • Target: Further FTC practice

Find the average value of $f(x) = \sin x$ on $[0, \pi]$, to three decimal places.



11 [Intermediate] • Target: Mean Value Theorem for integrals

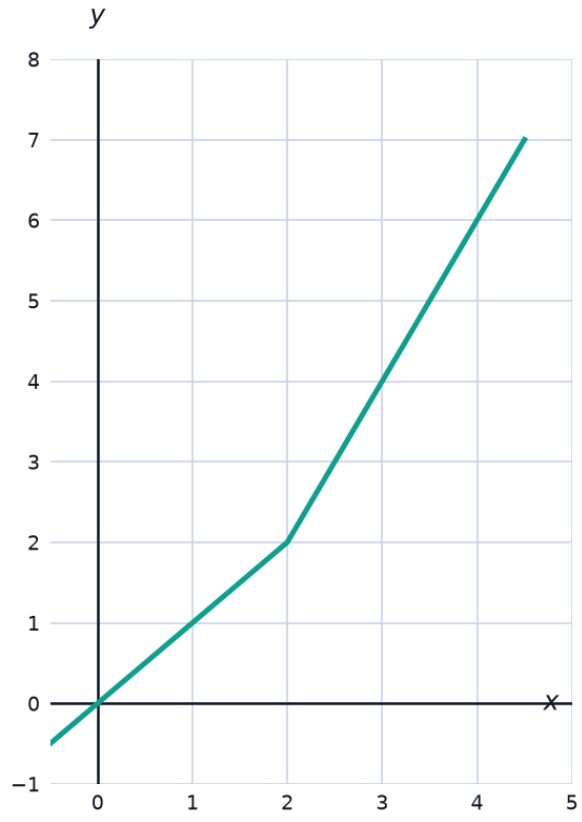
Find the value(s) of c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$ (where $f(c)$ equals the average value of f).

12 [Intermediate] • Target: Mean Value Theorem for integrals

Evaluate $\int_2^5 f(x)\,dx$ given that the average value of f on $[2, 5]$ is 7.

13 [Intermediate] • Target: Integrals of piecewise functions

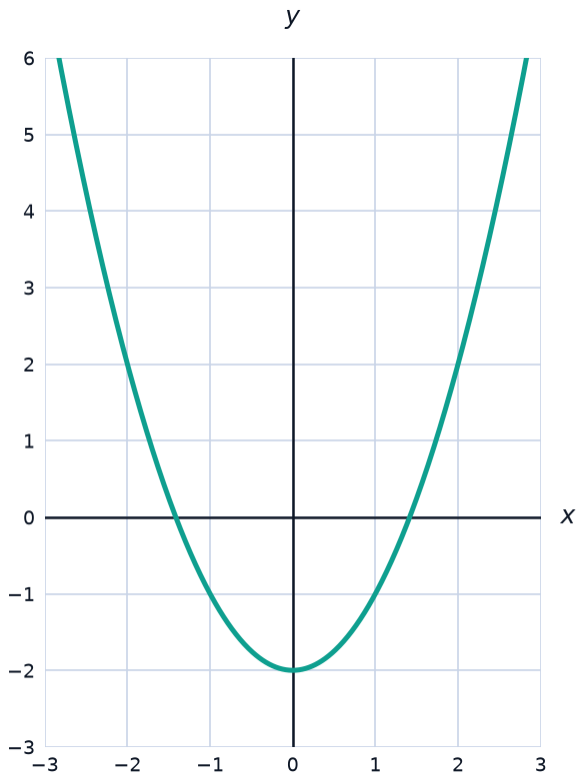
Evaluate $\int_0^4 f(x)\,dx$ where $f(x) = x$ for $x \le 2$ and $f(x) = 2x - 2$ for $x > 2$.



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14 [Intermediate] • Target: Visualizing the two parts of FTC together

For $g(x) = \int_0^x (t^2 - 2)\,dt$, find $g'(x)$ using FTC Part 1, then find where g has a local minimum by analyzing the sign of $g'(x)$.

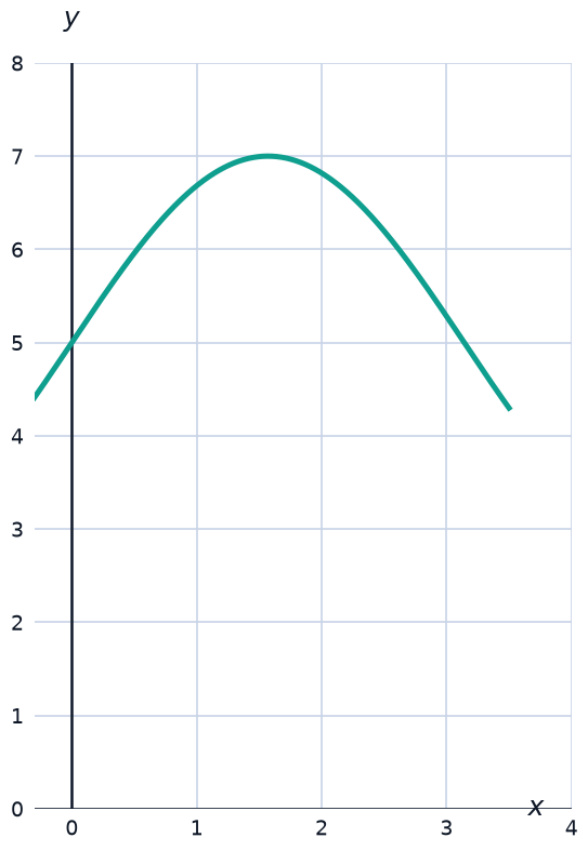


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: A capstone FTC application • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

The rate of water flowing into a tank is $r(t) = 5 + 2\sin(t)$ liters/minute. Find the total volume that flows in during $[0, \pi]$ minutes, using FTC.



16 [Challenging] • Target: FTC Part 1 with more complex upper limits • Cross-Lesson: 15.01 Double Integrals

Find $\frac{d}{dx}\int_0^{x^3} \sqrt{1+t^2}\,dt$ using the chain rule with FTC.

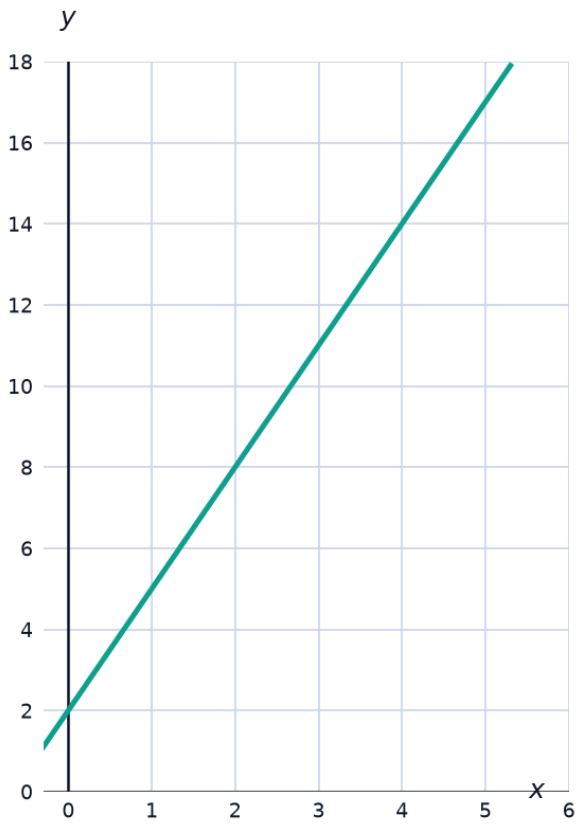
17 [Challenging] • Target: FTC Part 1 with more complex upper limits • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Find $\frac{d}{dx}\int_{x^2}^{5} \cos(t)\,dt$, using $\int_{x^2}^{5} = -\int_{5}^{x^2}$ before differentiating.

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18 [Challenging] • Target: Net change and the FTC • Cross-Lesson: 2.01 Evaluating Limits

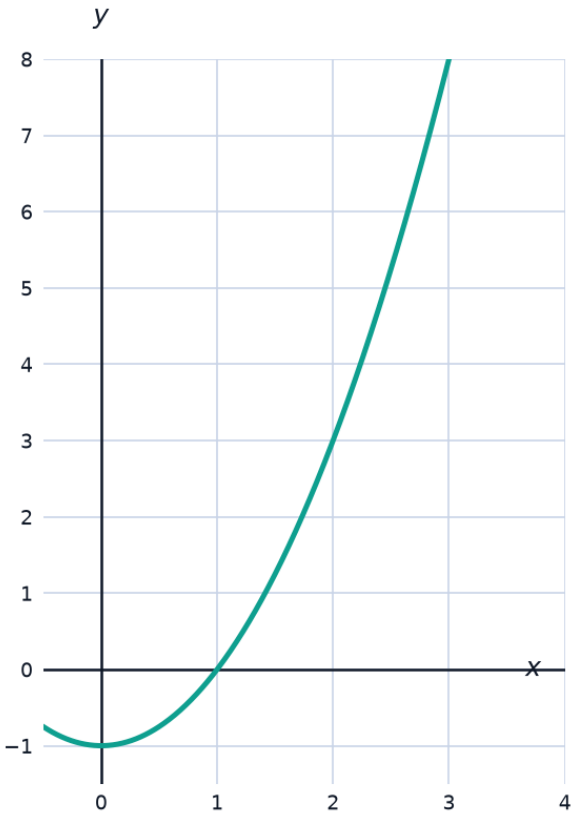
A tank starts with 200 liters of water. Water flows out at rate $r(t) = 3t + 2$ liters/minute for the first 5 minutes. Use FTC to find the amount of water in the tank at $t = 5$.



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19 [Challenging] • Target: Visualizing FTC Part 2 as area • Cross-Lesson: 2.02 Continuity

The graph shows $f(x) = x^2 - 1$ on $[0, 3]$. Use FTC to find $\int_0^3 (x^2 - 1)\,dx$ and explain why the result is not simply the area of the shaded region (part of the curve dips below the axis).



20 [Challenging] • Target: Combining FTC Part 1 and Part 2 in one problem • Cross-Lesson: 3.01 The Derivative from First Principles

Let $F(x) = \int_1^x (3t^2 - 2)\,dt$. Find $F(2)$ using Part 2, then verify $F'(2)$ using Part 1 and confirm it matches direct substitution into the integrand.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: FTC Part 1 (Derivative of an Accumulation Function)

If $g(x) = \int_1^x \sqrt{t^3 + 1}\,dt$, find $g'(x)$.

a. $\sqrt{x^3 + 1}$

b. $\frac{3x^2}{2\sqrt{x^3 + 1}}$

c. $\sqrt{x^3 + 1} - \sqrt{2}$

d. $\frac{1}{3}(x^3 + 1)^{3/2}$

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22 [Foundational] • Target: FTC Part 1 (Derivative of an Accumulation Function)

Find $\frac{d}{dx}\left[\int_x^5 \cos(t^2)\,dt\right]$.

a. $-\cos(x^2)$

b. $\cos(x^2)$

c. $-2x\sin(x^2)$

d. $\cos(25)-\cos(x^2)$

23 [Foundational] • Target: FTC Part 1 (Derivative of an Accumulation Function)

Find $h'(x)$ for $h(x)=\int_0^{x^2}\sin(t)\,dt$.

a. $2x\sin(x^2)$

b. $\sin(x^2)$

c. $2x\cos(x^2)$

d. $\sin(x^4)$

24 [Foundational] • Target: FTC Part 1 with Variable Limits on Both Bounds

Find $F'(x)$ for $F(x)=\int_{2x}^{3x}(t^2+1)\,dt$.

a. $27x^2+3-2(4x^2+1)=19x^2+1$

b. $19x^2+1$

c. $27x^2-8x^2=19x^2$

d. $5x^2+1$

25 [Foundational] • Target: FTC Part 1 with Variable Limits on Both Bounds

Evaluate $\frac{d}{dx}\left[\int_{\sin x}^{\cos x} e^t\,dt\right]$.

a. $-e^{\cos x}\sin x-e^{\sin x}\cos x$

b. $e^{\cos x}-e^{\sin x}$

c. $e^{\cos x}\sin x+e^{\sin x}\cos x$

d. $-e^{\cos x}+e^{\sin x}$

26 [Foundational] • Target: FTC Part 1 with Variable Limits on Both Bounds

Find the inflection points of $g(x)=\int_0^x(t^3-3t)\,dt$.

a. $x=1$ and $x=-1$

b. $x=0$ and $x=\pm\sqrt{3}$

c. $x=3$

d. No inflection points

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27 [Foundational] • Target: FTC Part 2 (Evaluating Definite Integrals)

Evaluate $\int_1^3 (3x^2 - 4x + 1)\,dx$.

a. 12

b. 14

c. 10

d. 16

28 [Foundational] • Target: FTC Part 2 (Evaluating Definite Integrals)

Evaluate $\int_0^{\pi/4} \sec^2(x)\,dx$.

a. 1

b. $\sqrt{2}$

c. 1/2

d. 0

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: FTC Part 2 (Evaluating Definite Integrals)

Evaluate $\int_1^e \frac{4}{x}\,dx$.

a. 4

b. $4e$

c. $4\ln(e) - 4 = 0$

d. 1

30 [Intermediate] • Target: Average Value of a Function

State the formula for the average value f_{avg} of a continuous function f on $[a, b]$.

a. $f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$

b. $f_{\text{avg}} = \frac{f(a) + f(b)}{2}$

c. $f_{\text{avg}} = \int_a^b f(x)\,dx$

d. $f_{\text{avg}} = \frac{f(b) - f(a)}{b-a}$

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31 [Intermediate] • Target: Average Value of a Function

Find the average value of $f(x) = 3x^2 - 2x$ on the interval $[0, 2]$.

a. 2

b. 4

c. 3

d. 1

32 [Intermediate] • Target: Average Value of a Function

By the Mean Value Theorem for Integrals, find $c \in [0, 3]$ such that $f(c) = f_{\text{avg}}$ for $f(x) = 2x$.

a. 1.5

b. 1.0

c. 2.0

d. 0.75

33 [Intermediate] • Target: FTC Part 1 (Derivative of an Accumulation Function)

For $F(x) = \int_a^x f(t)dt$ with f continuous, the proof of FTC Part 1 uses the Mean Value Theorem for Integrals to write $F(x + h) - F(x)$ as:

a. hf'(x)

b. $f(c) \cdot h$ for some $c \in [x, x + h]$

c. $f(x) \cdot (x + h)$

d. f(x+h)-f(x)

34 [Intermediate] • Target: FTC Part 1 with Variable Limits on Both Bounds

If f has a jump discontinuity at $c \in (a, b)$, the accumulation function $F(x) = \int_a^x f(t)dt$ is still:

a. continuous at c (though not differentiable there)

b. discontinuous at c

c. differentiable at c

d. undefined at c

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35 [Intermediate] • Target: FTC Part 2 (Evaluating Definite Integrals)

FTC Part 1: $G(x) = \int_0^x \sin(t^2)dt$. $G'(x) =$

a. $\sin(x^2)$

b. $\cos(x^2)$

c. $2x\sin(x^2)$

d. $\sin x^2/x$

36 [Intermediate] • Target: Average Value of a Function

FTC Part 2: $\int_a^b f(x)dx = F(b) - F(a)$ where $F' = f$. Evaluate $\int_0^\pi \cos x\,dx$:

a. 0

b. 2

c. -2

d. 1

37 [Intermediate] • Target: Part 2: evaluating definite integrals

Net change theorem: $\int_1^4 v(t)dt$ gives the net displacement if $v(t)$ is velocity. For $v = 2t$:

a. 15

b. 8

c. 30

d. 4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Part 1: differentiating an integral • Cross-Lesson: 2.01 Evaluating Limits

If f is continuous on $[a, b]$ and F is any antiderivative, FTC Part 2 requires no assumption that F is:

a. equal to f

b. unique $--$ any antiderivative works, since they differ by a constant

c. bounded on $[a, b]$

d. differentiable everywhere

39 [Challenging] • Target: Average value and applications • Cross-Lesson: 2.02 Continuity

The Net Change Theorem $\int_a^b F'(x)dx = F(b) - F(a)$ requires F' to be:

a. continuous and positive

b. integrable on $[a, b]$

c. a polynomial

d. bounded and increasing

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40 [Challenging] • Target: Further FTC practice • Cross-Lesson: 3.01 The Derivative from First Principles

FTC Part 1 shows every continuous function has an:

a. antiderivative

b. derivative everywhere

c. limit

d. inflection point

41 [Challenging] • Target: Mean Value Theorem for integrals • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Mean value theorem for integrals: $\int_a^b f\,dx = f(c)(b-a)$ for some $c \in (a, b)$. Requires f to be:

a. positive

b. integrable and bounded

c. continuous

d. analytic

42 [Challenging] • Target: Integrals of piecewise functions • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Let $F(x) = \int_2^x (t^2 - 1)dt$. According to the Fundamental Theorem of Calculus, what is the derivative of $F(x)$ with respect to x?

a. $x^2 - 1$

b. x^2

c. $x^2 - 1 - (2^2 - 1)$

d. $2x$

43 [Challenging] • Target: Visualizing the two parts of FTC together • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Mean Value Theorem for Integrals: if f is continuous on $[a, b]$, there exists $c \in (a, b)$ with $\int_a^b f(x)\,dx =$

a. $f(c)(b-a)$

b. $f'(c)(b-a)$

c. $f(b) - f(a)$

d. $f(c)$

44 [Challenging] • Target: A capstone FTC application • Cross-Lesson: 6.01 Related Rates

Find the average value of $f(x) = \cos(x)$ on $[0, \frac{\pi}{2}]$.

a. $\frac{2}{\pi}$

b. $\frac{4}{\pi}$

c. 1

d. $\frac{1}{\pi}$

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45 [Challenging] • Target: FTC Part 1 with more complex upper limits • Cross-Lesson: 6.02 Optimization and Curve Sketching

Net area from 0 to 2π of $\sin x$:

a. -2

b. 0

c. 2

d. 4

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